

Ledger/Reconciliation

#	Bài toán cụ thể	Function / journey	Data signal	AI/DS/ML	Output	Được gì	Rủi ro / mất gì
1	Double-entry posting validator	Khi tạo ledger transaction	debit lines, credit lines, amount, currency, account type	Rule + anomaly	valid/invalid posting	Học nền tảng ledger, rất fintech core	AI/ML nhẹ nếu đứng riêng
2	Debit-credit imbalance detection	Journal/posting validation	total debit, total credit, rounding, currency	Rule + DS anomaly	imbalance alert	Cực cụ thể, dễ demo	Quá đơn giản nếu chỉ check sum
3	Invalid account type posting detection	Ledger posting	account type: asset/liability/revenue/expense, transaction type	Rule + ML-lite	invalid account mapping	Học accounting mapping	Cần hiểu accounting cơ bản
4	Transaction-to-posting mapping checker	Business transaction → ledger	payment/refund/fee type, expected postings, actual postings	Rule + anomaly	missing/wrong posting	Học business transaction khác journal entry	Domain hơi khó

5	Wallet balance drift checker	Wallet balance validation	balance snapshot, postings, transaction history	DS/rule	balance mismatch	Hiểu vì sao không chỉ lưu balance	Cần thiết kế ledger data tốt
6	Balance reconstruction from postings	Ledger audit	account postings, opening balance, closing balance	DS/rule computation	reconstructed balance	Rất core, dễ giải thích	Ít AI nếu không thêm anomaly
7	Negative balance anomaly detection	Wallet/ledger monitoring	account balance, account type, allowed overdraft	Rule + anomaly	negative balance alert	Function nhỏ, thực tế	Có thể quá đơn giản
8	Suspicious manual adjustment detection	Backoffice adjustment	staff_id, amount, reason, account, time	Anomaly detection	suspicious adjustment alert	Có audit/risk angle mạnh	Dữ liệu synthetic, nhạy cảm
9	Reversal consistency checker	Refund/reversal lifecycle	original transaction, reversal transaction, posting pairs	Rule + anomaly	invalid reversal alert	Học reverse transaction đúng cách	Domain phức tạp hơn CRUD
10	Refund posting validator	Refund flow	original payment, refund amount, prior refunds, postings	Rule + anomaly	valid/invalid refund posting	Rất thực tế trong payment/ledger	Có thể trùng payment ops nếu không rõ scope

11	Fee posting correctness checker	Payment fee/accounting	gross amount, fee amount, net settlement, fee account	Rule + anomaly	fee posting issue	Học gross/net/fee accounting	Cần mô phỏng fee tốt
12	Cashback/reward posting checker	Wallet rewards	payment, cashback, promotion account, user wallet	Rule + anomaly	cashback posting issue	Hợp ví điện tử/super app	Domain reward accounting hơi rối
13	Merchant payable balance checker	Merchant settlement ledger	merchant payments, refunds, fees, payouts	DS/rule	payable balance mismatch	Rất thực tế merchant payment	Cần mô phỏng merchant lifecycle
14	Provider clearing account balance checker	Payment provider clearing	provider payments, settlements, refunds, fees	DS/reconciliation	clearing balance issue	Học clearing account/internal ledger	Domain khó hơn wallet đơn giản
15	Internal-vs-provider transaction matching	Reconciliation	internal payment_id, provider reference, amount, date, merchant	DS matching/fuzzy matching	matched/unmatched	Rất thực tế fintech backend	Data synthetic phải đủ đa dạng
16	Missing provider settlement detection	Settlement recon	internal success payment, provider statement missing	Rule + anomaly	missing settlement alert	Cụ thể, dễ hiểu	Cần mô phỏng delay/timing

17	Unexpected provider transaction detection	Settlement recon	provider record không có internal payment	Anomaly/rule	unexpected external record	Học reconciliation two-way	Cần xử lý false positive do timing
18	Amount mismatch detection	Internal vs provider	internal amount, provider amount, currency, fee, refund	Rule + anomaly	amount mismatch type	Rất thực tế, dễ demo	Có thể rule-heavy
19	Fee/net settlement mismatch classifier	Provider settlement	gross, fee, net amount, expected fee schedule	ML-lite/rule classification	fee mismatch reason	Gắn DS/ML hơn amount check thường	Cần label synthetic
20	Currency/rounding mismatch detector	Multi-currency settlement	currency, FX rate, rounding rules, provider amount	Rule + anomaly	FX/rounding mismatch	Có domain depth	Có thể quá khó nếu thêm FX
21	Refund reconciliation mismatch	Refund settlement	original payment, refund record, provider refund, bank statement	DS matching/rule	refund matched/mismatch	Practical và cụ thể	Refund lifecycle dễ phức tạp
22	Partial refund reconciliation	Partial refund flow	original amount, multiple refund records, settlement amount	Rule + DS	partial refund status	Học nhiều edge cases	Scope dễ phình

23	Chargeback reconciliation	Dispute/chargeback	chargeback record, original payment, merchant debit	DS/rule	chargeback matched/missing	Học payment dispute accounting	Dataset khó
24	Payout reconciliation	Merchant payout	merchant payable, payout instruction, bank statement	DS matching	payout matched/unmatched	Rất gần merchant ops	Cần tạo payout statement
25	Payout delay detection	Merchant settlement	payout due date, actual payout date, bank/provider status	Anomaly/time-series	delayed payout alert	Có ops value rõ	Dễ sang payment ops
26	Settlement batch completeness check	Daily settlement batch	expected transactions, batch file records, totals	Rule + DS	batch complete/incomplete	Rất thực tế batch processing	AI/ML nhẹ
27	Batch total control check	Settlement batch validation	record count, total amount, merchant totals, provider totals	Rule + anomaly	control total mismatch	Học reconciliation controls	Quá đơn giản nếu đứng riêng
28	Duplicate settlement record detection	Provider/bank statement	ref_id, amount, date, merchant, repeated rows	Dedup/entity matching	duplicate settlement record	Cụ thể, dễ evaluate	Có thể giống data cleaning

29	Near-duplicate transaction matching	Fuzzy reconciliation	amount close, date close, merchant alias, reference typo	Fuzzy matching/ML	candidate match score	Có ML/DS matching rõ	Cần design matching metric
30	Reconciliation confidence scoring	Matching engine output	exact/fuzzy match features, amount/date distance	ML classification/ranking	match confidence	Rất hay, biến rule matching thành ML	Cần label synthetic
31	Auto-match vs manual-review routing	Recon exception workflow	match confidence, mismatch type, amount materiality	ML/rule decision	auto-match/review route	Human-in-the-loop rõ	Cần threshold tuning
32	Reconciliation exception classification	Recon ops	mismatch fields, transaction type, provider, timing	ML classification	exception type	Có ML rõ, practical	Cần label exception types
33	Exception severity scoring	Recon exception queue	amount, age, merchant, mismatch type, frequency	Ranking/rule + ML-lite	low/medium/high severity	Hợp ops prioritization	Subjective nếu không có business rules
34	Aging exception monitor	Unresolved recon items	exception age, amount, owner, status	DS analytics	aging report	Thực tế trong ops	Ít ML nếu đứng riêng

35	Reconciliation root-cause assistant	Ops review	internal record, provider record, mismatch type, history	LLM structured explanation + rules	possible root cause	Hợp AI assistant nếu grounded	Không nên để LLM đoán vô căn cứ
36	Reconciliation report generator	End-of-day recon	matched count, unmatched, amount mismatches, aging exceptions	DS + LLM summary	daily recon report	Hợp AI + DS dashboard	Nếu chỉ report text thì nhẹ
37	Settlement anomaly dashboard	Ops monitoring	mismatch rate, failed match rate, provider, merchant	DS anomaly	provider/merchant anomaly	Học monitoring mindset	Cần time-series synthetic
38	Merchant-level reconciliation quality score	Merchant ops	match rate, mismatch rate, refund issue rate, payout delay	DS scoring	merchant recon quality	B2B/payment angle hay	Cần nhiều merchant data
39	Provider-level reconciliation quality score	Provider/PSP ops	settlement delay, mismatch frequency, missing records	DS scoring/anomaly	provider quality score	Hợp payment platform ops	Dữ liệu provider synthetic
40	Ledger-to-bank statement reconciliation	Bank account reconciliation	ledger cash account postings, bank statement lines	DS matching	matched/unmatched bank lines	Rất thực tế accounting	Cần mô phỏng bank statement

41	Cash account balance reconciliation	Ledger vs bank balance	ledger cash balance, bank closing balance, pending items	Rule + DS	cash balance difference	Học bank reconciliation	Có thể nặng accounting
42	Pending settlement account monitor	Clearing/pending account	pending items, age, expected settlement date	DS/anomaly	stale pending item	Rất fintech/payment practical	Cần hiểu clearing/pending
43	Aging pending balance detector	Ledger monitor	pending account balance by age bucket	DS/anomaly	aging balance alert	Học ops finance controls	Ít ML nếu đơn giản
44	Revenue recognition mismatch	Fee/revenue ledger	fee events, revenue postings, settlement fee records	Rule + anomaly	revenue mismatch	Có accounting depth	Có thể quá kế toán
45	Charge/fee leakage detection	Payment fee accounting	expected fee, actual posted fee, merchant pricing rule	DS/anomaly	fee leakage alert	Business value rõ	Need pricing rules synthetic
46	Unposted transaction detector	Event-to-ledger pipeline	successful payment event, missing ledger posting	Rule + anomaly	unposted transaction alert	Rất thực tế event/ledger consistency	Cần event pipeline giả lập

47	Posted-without-event detector	Ledger audit	ledger posting không có business event	Anomaly/rule	orphan posting alert	Học audit consistency	Cần event sourcing mindset
48	Event-ledger consistency checker	Business events vs postings	event type, expected posting template, actual posting	Rule + anomaly	consistency report	Rất hay cho backend fintech	Domain mapping cần làm kỹ
49	Journal entry explanation assistant	Reviewer xem journal entry	transaction type, postings, accounts, amounts	LLM grounded explanation	human-readable explanation	Hợp học accounting/journal	Không nên để LLM tự suy luận sai
50	Transaction-to-journal template recommender	New product/config	transaction type, account mappings, product config	Rule/ranking/LLM suggestion	suggested posting template	Hay nhưng khó, gần accounting config	Dễ over-scope
51	Financial product GL mapping validator	Product setup	product type, GL accounts, fee/interest mappings	Rule + anomaly	mapping issue	Học Fineract/product config angle	Nặng domain core banking
52	Loan repayment allocation validator	Loan repayment accounting	principal, interest, fee, penalty, postings	Rule + anomaly	allocation/posting mismatch	Học lending + accounting	Domain khó

53	Savings interest posting checker	Savings account interest	balance, interest rate, posting schedule, journal	Rule + anomaly	interest posting issue	Học core banking sâu	Có thể quá ngân hàng lỗi
54	Daily close reconciliation checker	End-of-day close	ledger totals, pending postings, failed batches, exceptions	DS/rule	close readiness status	Học COB/batch mindset	Scope lớn nếu làm full
55	Settlement delay predictor	Expected settlement	provider, method, bank, day, amount, holidays giả lập	ML/time-series	expected settlement date	Có ML rõ, gần payment/recon	Cần label synthetic
56	Recon workload prediction	Ops planning	daily transaction volume, mismatch rate, exception age	Forecasting/DS	expected workload	Hợp ops analytics	Không quá fintech nếu thiếu recon details
57	Manual adjustment explanation checker	Backoffice audit	adjustment reason, amount, account, supporting event	NLP/LLM + rule	sufficient/insufficient explanation	Có compliance angle	Nhạy cảm, cần synthetic
58	Manual adjustment approval routing	Maker-checker workflow	amount, account type, staff role, adjustment reason	Rule/ML risk scoring	require approval level	Học maker-checker trong ledger	AI/ML nhẹ nếu chỉ rule

59	Reconciliation case similarity search	Ops tìm case tương tự	exception features, historical resolution, notes	Embedding/vector search	similar past cases	Hợp AI search, practical	Cần case history synthetic
60	Recon policy RAG assistant	Ops hỏi quy trình xử lý	SOP docs, exception facts	RAG + citation/eval	grounded answer	Hợp AI ops assistant	Chỉ tốt nếu có policy KB rõ

Package 1 — Merchant Settlement & Reconciliation Assistant

Gồm:

#15 Internal-vs-provider transaction matching
 #18 Amount mismatch detection
 #21 Refund reconciliation mismatch
 #30 Reconciliation confidence scoring
 #32 Exception classification
 #36 Reconciliation report generator

Package 2 — Wallet Ledger Integrity Checker

Gồm:

#1 Double-entry posting validator
 #4 Transaction-to-posting mapping checker
 #5 Wallet balance drift checker
 #9 Reversal consistency checker
 #46 Unposted transaction detector
 #48 Event-ledger consistency checker

Package 3 — Reconciliation Ops Intelligence

Gồm:

- #31 Auto-match vs manual-review routing
- #32 Exception classification
- #33 Exception severity scoring
- #34 Aging exception monitor
- #35 Root-cause assistant
- #59 Case similarity search

Package 4 — Provider / Merchant Settlement Quality Monitoring

Gồm:

- #37 Settlement anomaly dashboard
- #38 Merchant-level reconciliation quality score
- #39 Provider-level reconciliation quality score
- #42 Pending settlement account monitor
- #55 Settlement delay predictor

Package 5 — Core Banking Accounting Validator

Gồm:

- #49 Journal entry explanation assistant
- #51 Financial product GL mapping validator
- #52 Loan repayment allocation validator
- #53 Savings interest posting checker
- #54 Daily close reconciliation checker

